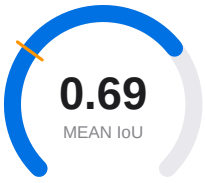
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419151561 is the best physical fit.

Forward replay tested 5 candidates; the runner-up fits at 0.52. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.69	**5** / 6	**79%**	**5**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **5 of 6** forcing variants
- Ahead of the runner-up by a mean loss gap of **0.1058**
- Inside the origin area **24 h** before the image
- The nearest vessel, **419765325**, 0.0064° away, is not supported

Candidates, ranked by physical fit

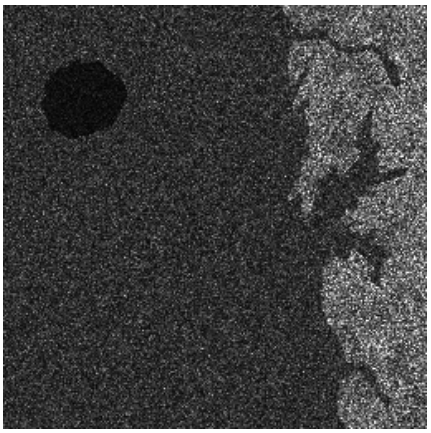
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419151561		0.692	5 of 6	9 Sep 2026, 22:30 UTC
2	419133241		0.522	0 of 6	9 Sep 2026, 16:30 UTC
3	419765325		0.212	1 of 6	10 Sep 2026, 04:00 UTC
4	419411265		0.000	0 of 6	9 Sep 2026, 23:30 UTC
5	419932010		0.000	0 of 6	9 Sep 2026, 23:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419765325**, ranked 3 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

79%
LEAD RETAINED

80 tested changes to the model's assumptions; **17** changed the leader.

The smallest change that flips it: **diffusivity multiplier 0.8** (magnitude 0.25), under which **419133241** leads.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	479 km^2	0%	44%	0%
+12 h 10 Sep 2026, 16:30 UTC	538 km^2	0%	41%	0%
+24 h 11 Sep 2026, 04:30 UTC	586 km^2	0%	30%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

1	Unassessed	never inside where track data exists, but no track data at one or more queried times — missing evidence, NOT exoneration
5	Candidate	track data present and inside the backward envelope at some queried time

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**71%** 41/58	**81%** 57/70
False alerts	**29%** 25/86	**14%** 12/86
Mistaken promotions	**30%** 12/40	**63%** 19/30

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:10 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`050a959a6aa20246ba5fc3f890dc76d3614b92a011d4c1868f15f099a1edd3e0`
coverage_result.json	`85f1bed43a7df27f7b85871c878aa62930a32d384959ee0ec9d055ae04217dc9`
forcing.npz	`eae171037b53ea6bdf5a452a1ad34e1cab4f9f9179d0d99728c916d6ebe87cac`
forecast_result.json	`cc88cf20cdb428dc9a2fd9e85c6f71db0094d56dc154e9ddecf5bc06686d2824`
fragility_result.json	`6c785b6d07cb23c3efafa172419d42ec72a82469442078c3fb83822b622a3419`
historical_mapping.json	`dba9c53a2f75c92607882c78f1d1a9cee3112e71d29d2c7a99dcc3a253669945`
manifest.json	`568ff8095a1c50f48fca9f8c46cf1ce690ad27b5aeef1c05e373e9153ea7b8cb`
mask_groundtruth.png	`1b96d7a9eb6ffc726a0203627634c2fff1c9c23ac928c231a24431f65bbae18a`
naive_baseline_result.json	`3bc0364460a2bf0bd1b2cd19bdf653f196ee9282fae6f0e92d93f778e6ad4a9d`
next_observation_result.json	`7fb0238f9259a35537fc5ff738d7a2e3f01c646dbc141a7e79bb5626b4a430e7`
scene.png	`3e22f8d89506b897cd86ec252fa45d1afbf3f981217a373c255a084fe63d1e35`

Uncertainty

— Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

— Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

— Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

— Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.